

FIND – S ALGORITHM

CODE

```
from google.colab import files

import pandas as pd

import os

uploaded = files.upload()

filename = list(uploaded.keys())[0]

print("\nUploaded File:", filename)

try:

    if filename.endswith(".csv"):

        try:

            data = pd.read_csv(filename)

        except UnicodeDecodeError:

            try:

                data = pd.read_csv(filename, encoding="latin1")

            except:

                data = pd.read_csv(filename, encoding="cp1252")

    elif filename.endswith(".xlsx") or filename.endswith(".xls"):

        data = pd.read_excel(filename)

    else:

        raise Exception("Unsupported file format!")

except Exception as e:

    print("Error reading file:", e)

    exit()

print("\nTraining Dataset:\n")

print(data)
```

```

concepts = data.iloc[:, :-1].values.tolist()
target = data.iloc[:, -1].values.tolist()
hypothesis = concepts[0].copy()
print("\nInitial Hypothesis:")
print(hypothesis)
for i in range(len(concepts)):
    if str(target[i]).strip().lower() == "yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'
        print("\nHypothesis after Example", i + 1)
        print(hypothesis)
print("\n=====")
print("Final Hypothesis")
print("=====")
print(hypothesis)

```

OUTPUT

```

Training Dataset:
  Sky AirTemp Humidity Wind Water Forecast EnjoySport
0 Sunny Warm Normal Strong Warm Same Yes
1 Sunny Warm High Strong Warm Same Yes
2 Rainy Cold High Strong Warm Change No
3 Sunny Warm High Strong Cool Change Yes

Initial Hypothesis:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Hypothesis after Example 1
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Hypothesis after Example 2
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Hypothesis after Example 4
['Sunny', 'Warm', '?', 'Strong', '?', '?']

=====
Final Hypothesis
=====
['Sunny', 'Warm', '?', 'Strong', '?', '?']

```